Introduction to Biomaterials

Lecture [1]

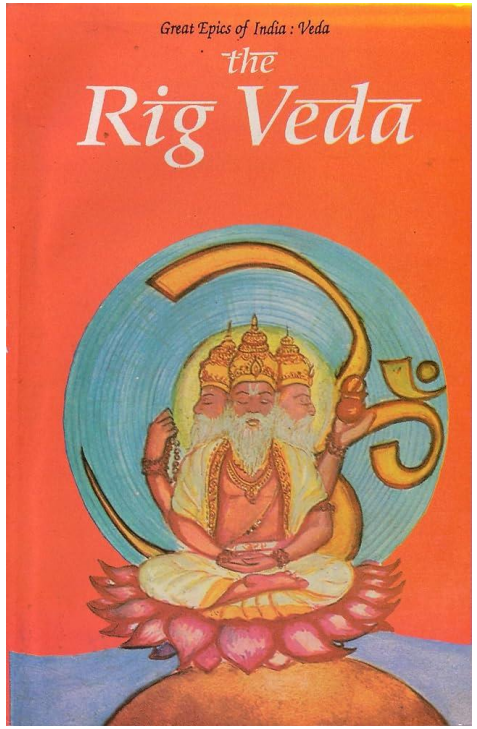
Overview of biomaterials: definition, types, properties, and medical applications

Goals

- After reading this chapter the student will understand the following.
  - History of implants and biomaterials.
  - Various definitions for biomaterials.
  - Different types of chemical bonds.
  - Basic families of materials.
  - Future directions for the progress in biomaterials.

Early Uses of Biomaterials

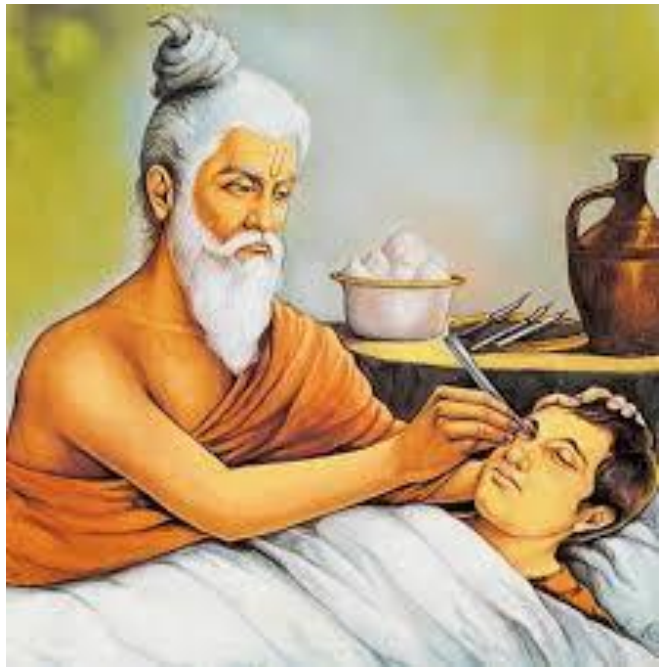
- Ancient texts like the **Rig Veda (over 3000 years old)** describe early prosthetics (e.g., Vishpla's iron leg) and surgical treatments (sutures for wounds).





Early Uses of Biomaterials

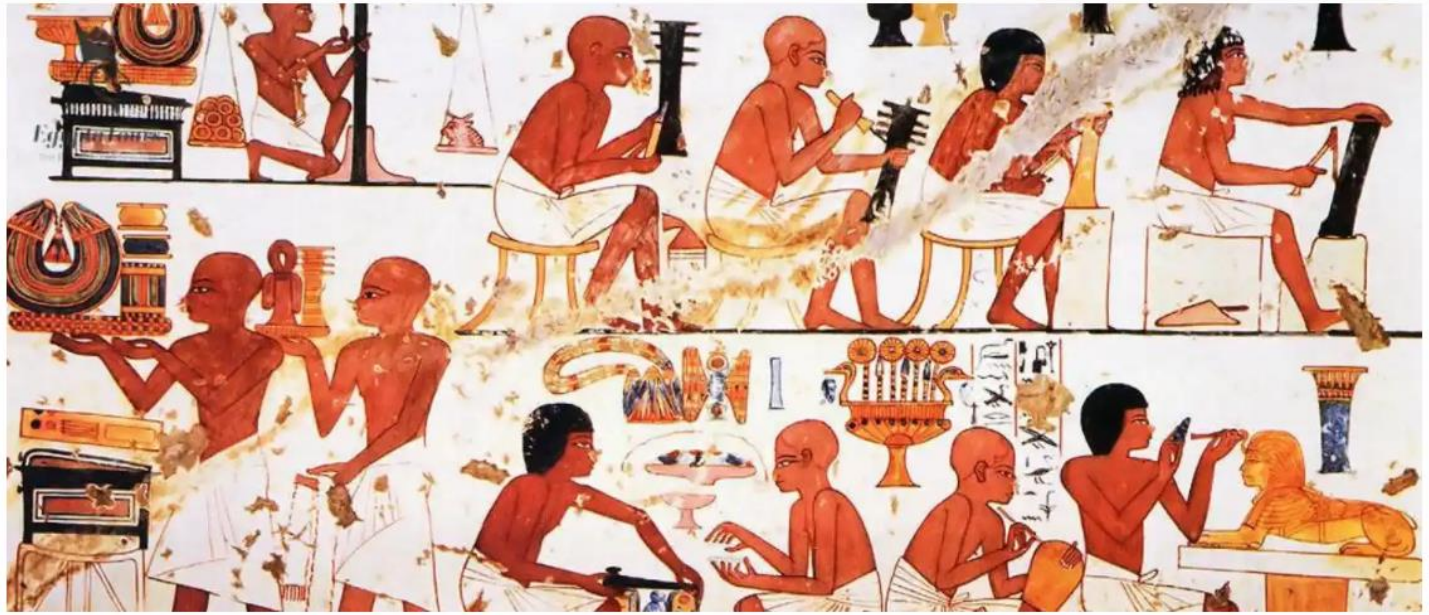
- Around **600 BC**, **Sushruta** documented advanced surgical techniques, including **nose reconstruction** and use of **natural sutures**.





Early Uses of Biomaterials

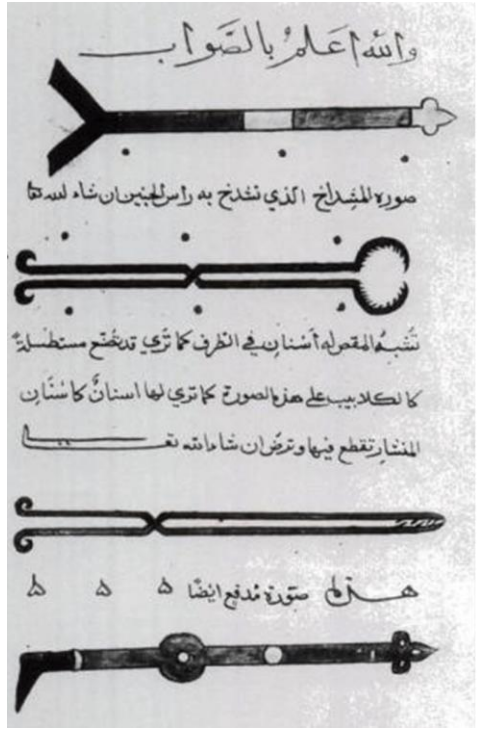
- Early sutures were made from **plant fibers, leather, tendons, and hair**; Egyptians used **linen sutures** ~4000 years ago.



Early Uses of Biomaterials

- Around the **10th century**, Al-Zahrawi introduced catgut sutures made from animal intestines, which could be absorbed by the body.

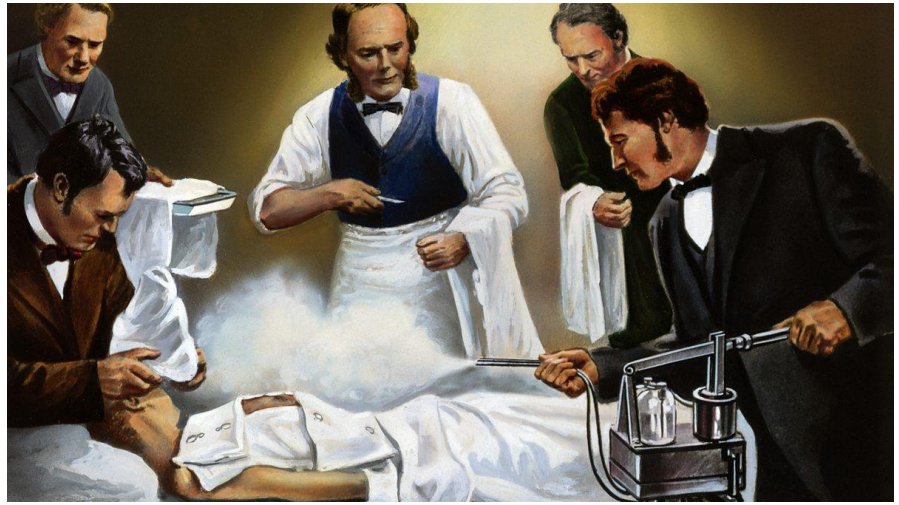




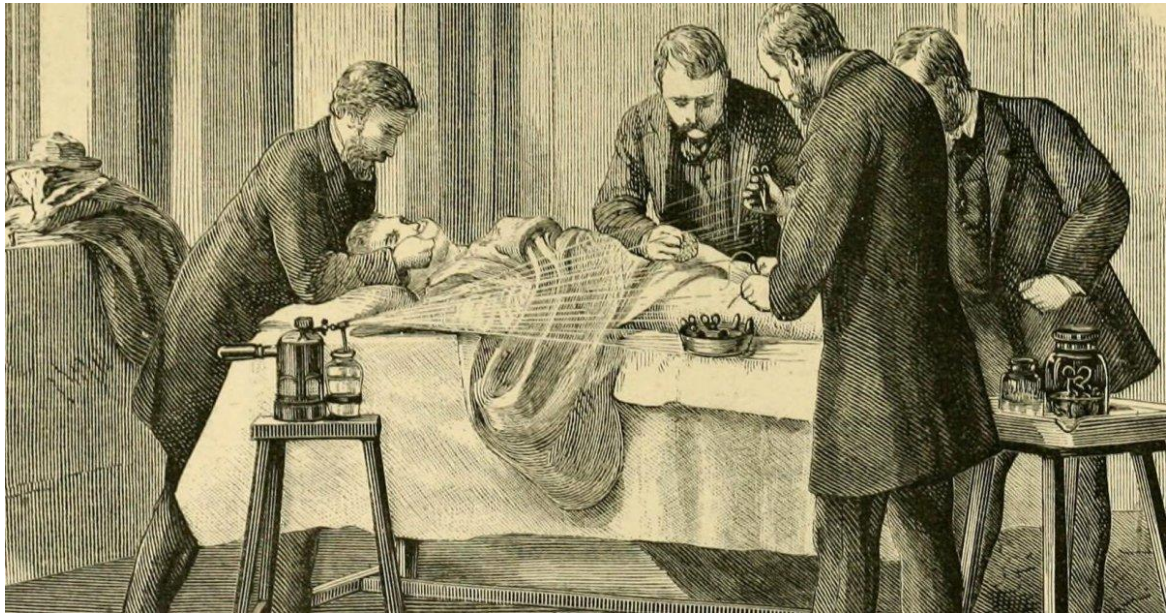


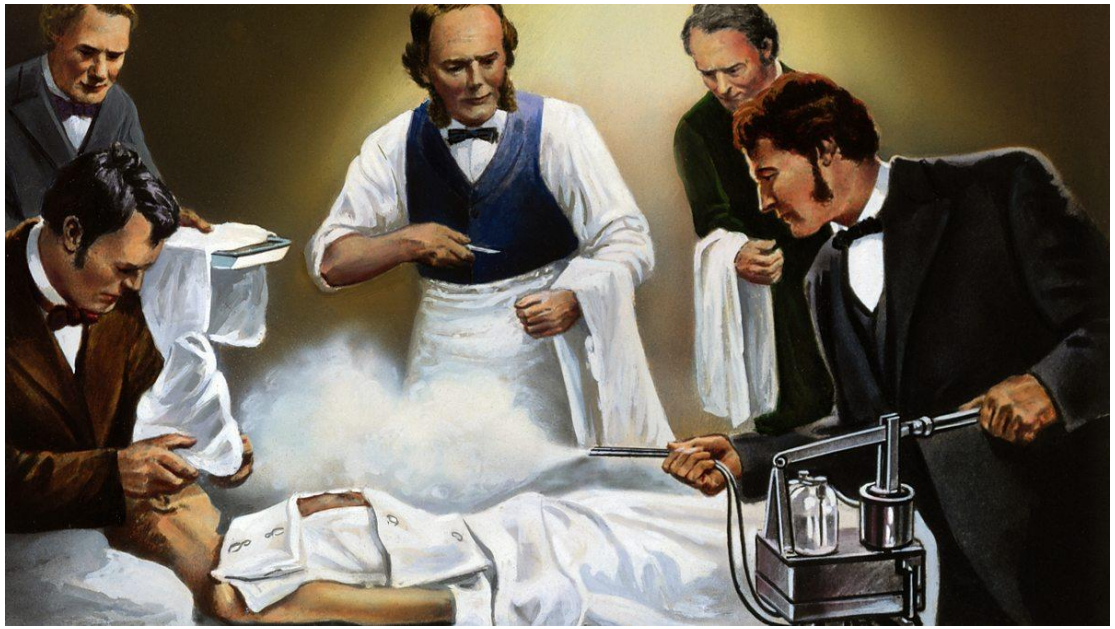
Early Uses of Biomaterials

- Humans have long tried to restore body function using external materials → **early biomaterials**.
- Early **implants inside the body often failed** due to infection.
- Major progress came with:
  - **Aseptic surgery (1860s, Lister)**
  - **Antibiotics (mid-1900s)**
- Today, implants are widely successful and improve/save millions of lives.

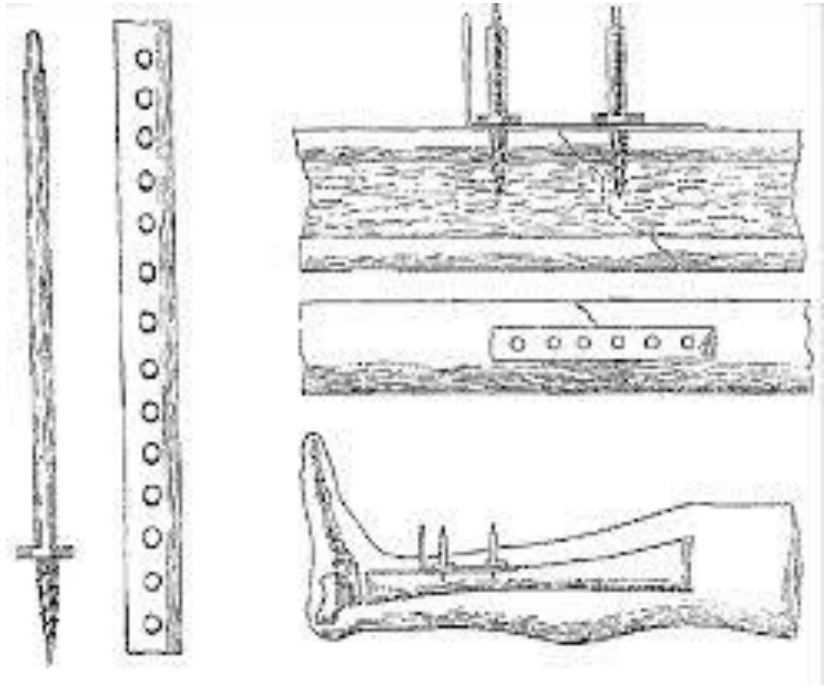


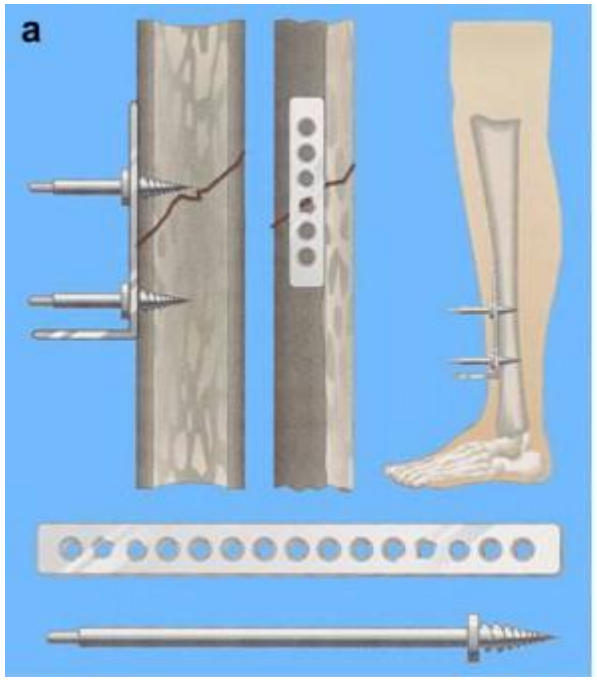
1860s: J. Lister- Aseptic surgical technique





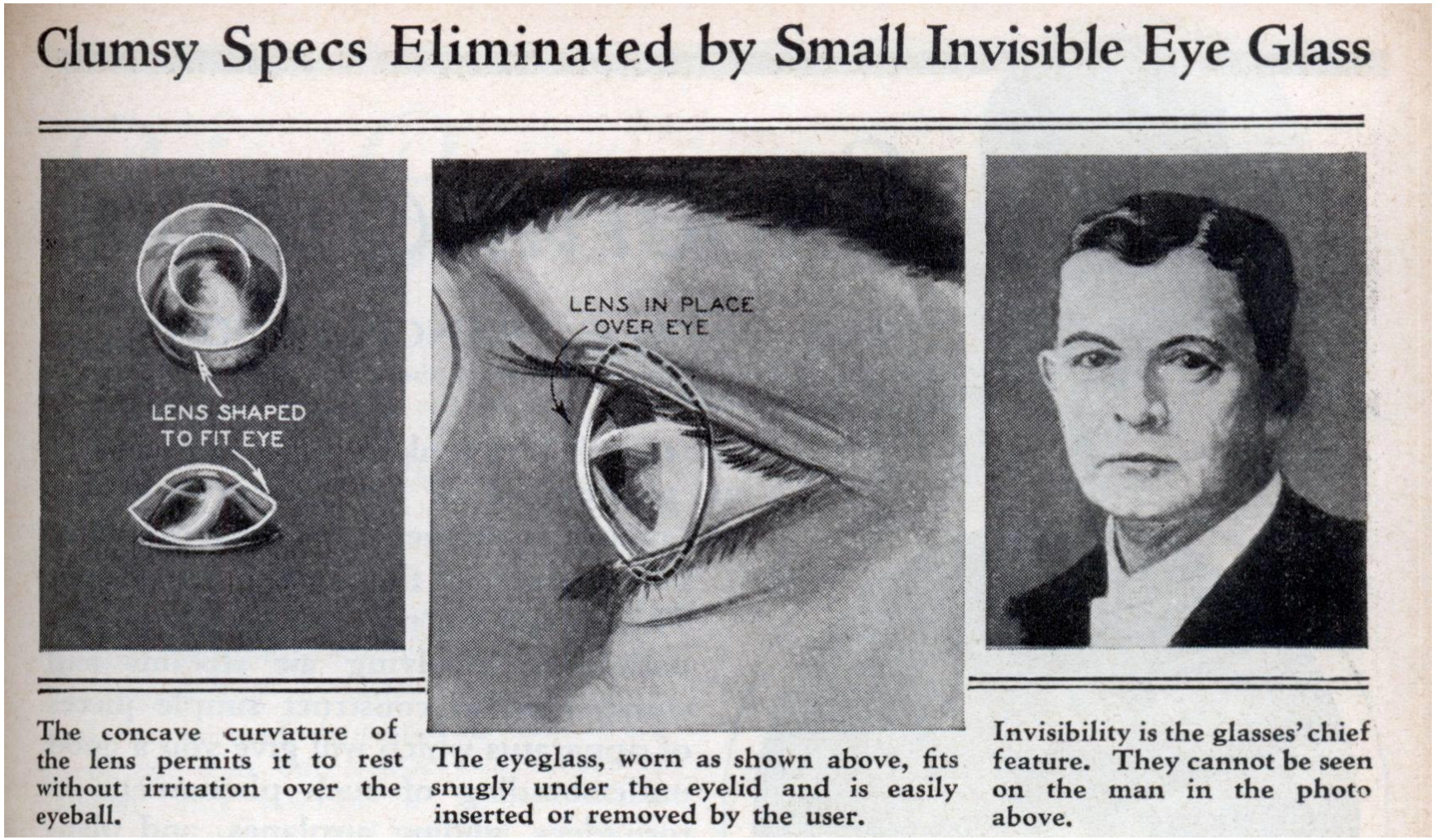
1886: C. Hansmann - Plates, screws for fracture fixation



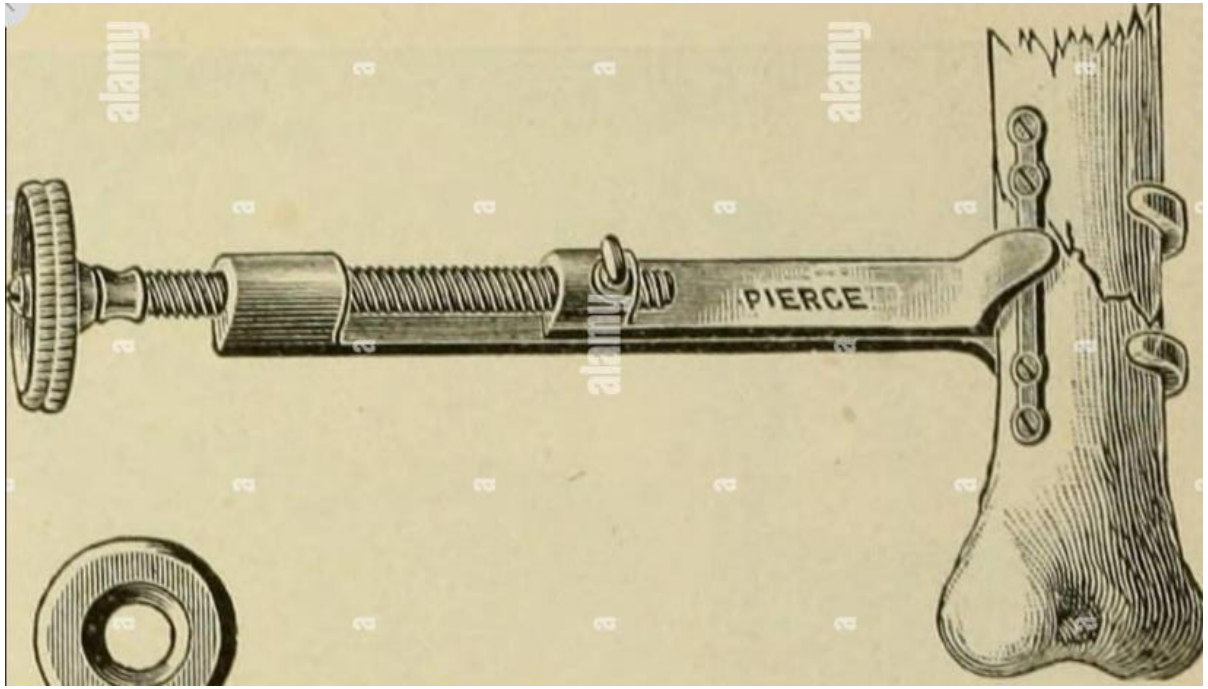




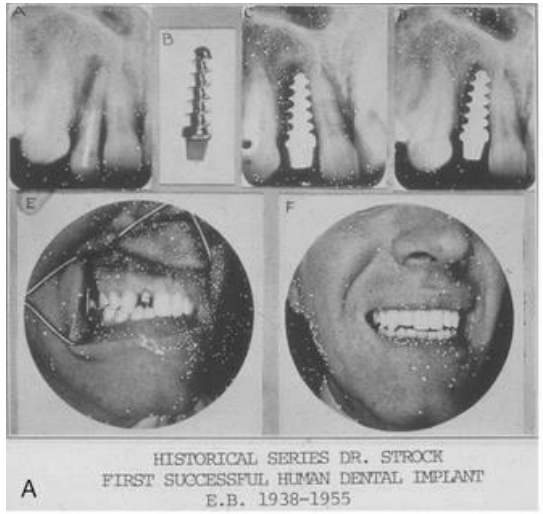
1887: Adolf Fick Glass, contact lens

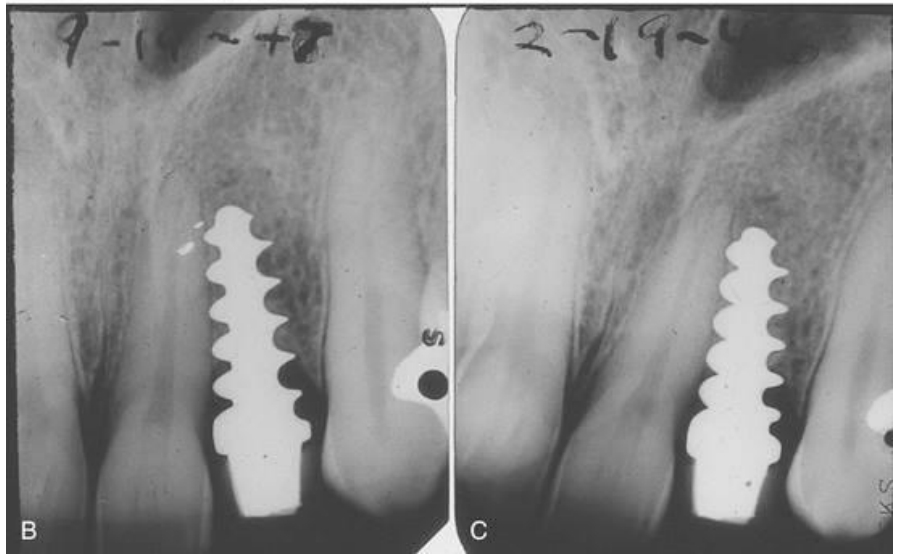


1912: W. D. Sherman - Vanadium steel plates and screws



1936–37: A. E. Strock - Vitallium for dental implants





1938: Philip Wiles - Total hip replacement



An Historical Perspective

The history of metal on metal articulation began at the Middlesex Hospital in London in 1938. Philip Wiles carried out total hip replacements in six patients with Still's disease [1]. The acetabular and femoral components were made of stainless steel and were ground together to ensure an accurate fit. The cup was fixed with two screws and the femoral component was secured by a bolt inserted down the femoral neck, attached to a plate on the lateral aspect of the femur (Fig. 11.1). Unfortunately all the radiographs were lost during the Second World War. Only one patient, with a disintegrated prosthesis, was still alive in 1951. In 1957 Wiles developed another design with better engineering and a more robust method of fixation. He performed eight operations, but the results were disappointing because of bone resorption and prosthetic loosening.

A similar stainless steel implant was used by Kenneth Coutts in Boston, USA [2]. He implanted one prosthesis that was revised 6 years postoperatively. In 1951 Edward Haboush at the Hospital for Joint Diseases in New York inserted a Vitallium total hip replacement with accurately matched articulating surfaces. Both components were fixed with Cranioplast®, a type of polymethylmethacrylate cement, which at the time was used for the repair of skull defects. The joint dislocated in the early postoperative period because of a vertically positioned acetabular component and the construct never actually functioned.

In the early 1950s Kenneth McKee, who had worked with Wiles as his Senior Registrar, implanted three metal on metal hip replacements based on his hip arthrodesis screw [3, 4]. Two were made from stainless steel and these became loose within 1 year. The

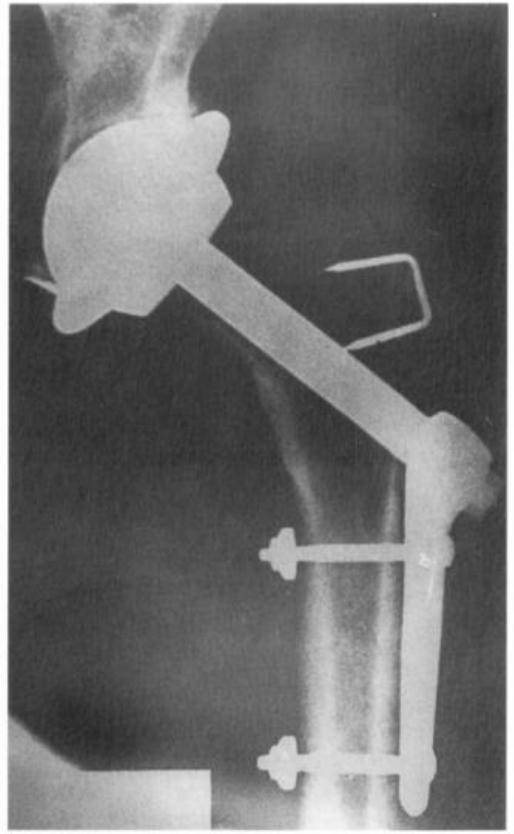
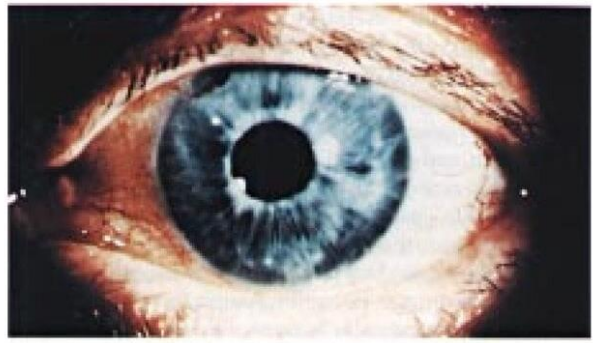


Fig. 11.1. The Wiles total hip replacement.

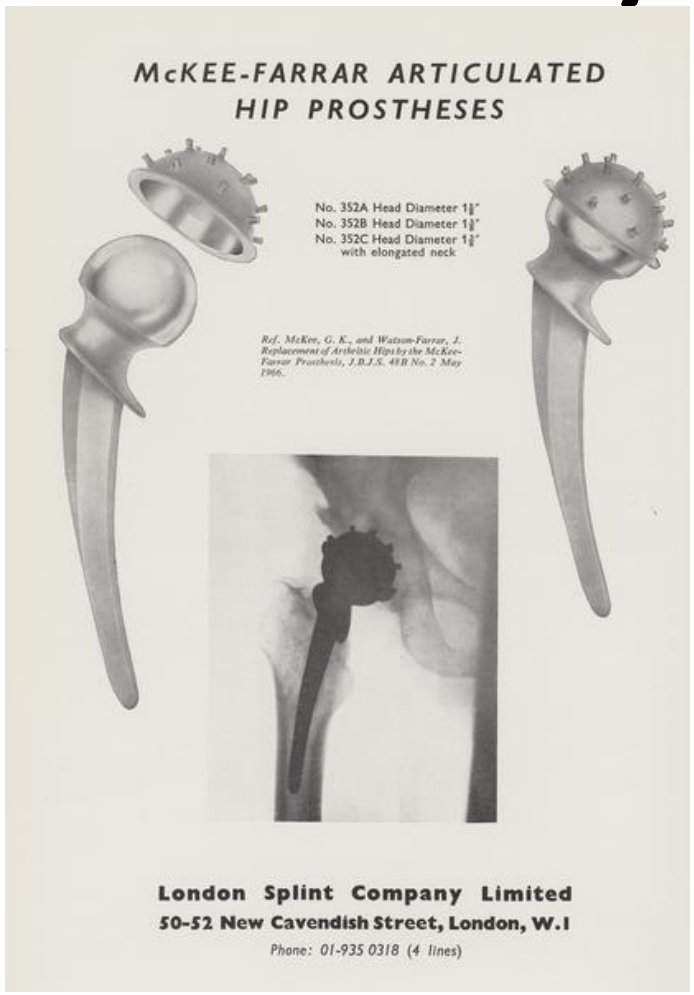
1949: Sir Harold Ridley - Intraocular lens







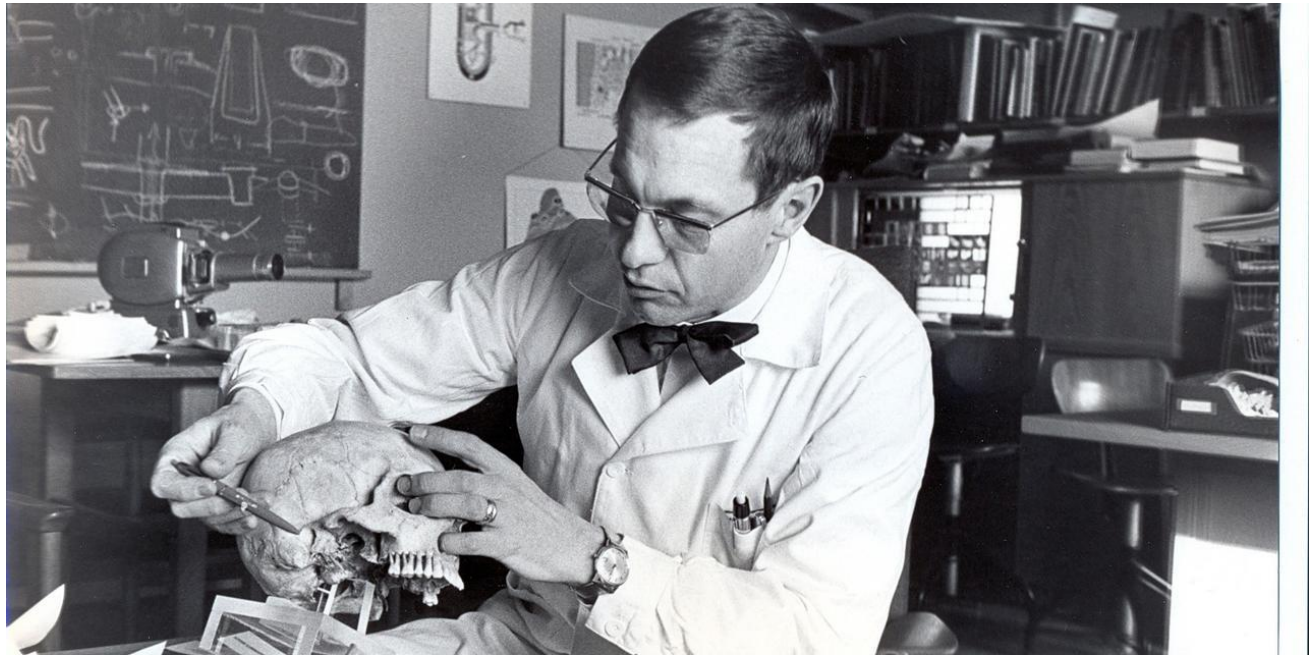
1951: G. K. McKee, J. Watson Farrar - Biomechanically successful total hip design

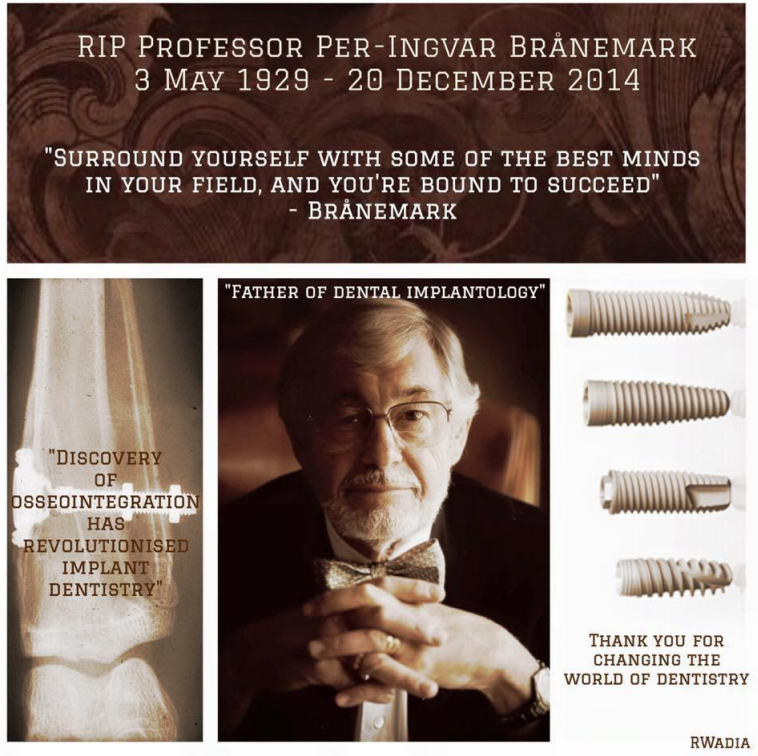






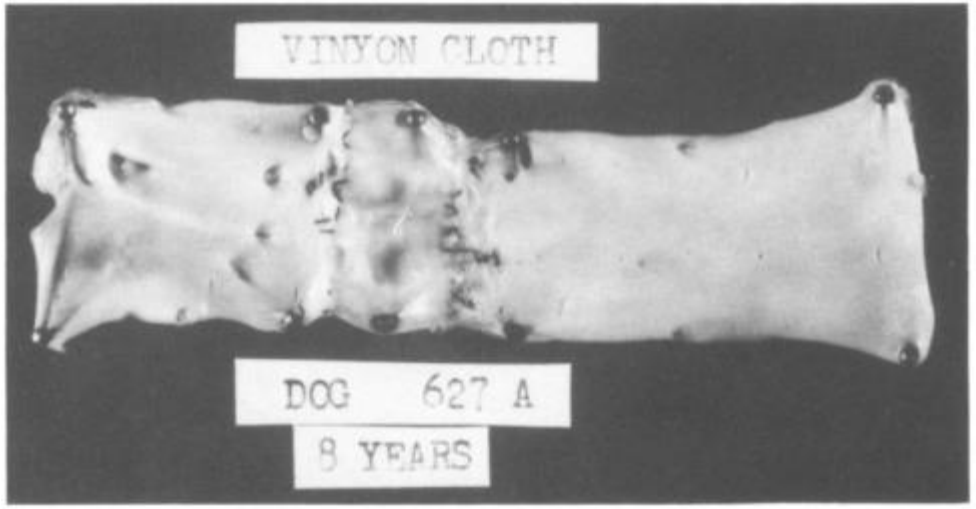
1952: Ingvar Branemark - Osteointegration of metal implants

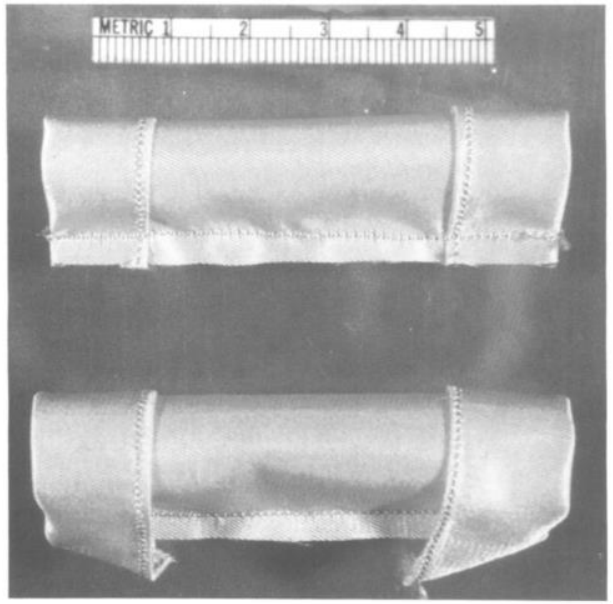




1952: A. B. Voorhees - Prosthetic vascular graft







1958: Earl E. Bakken - Wearable pacemaker

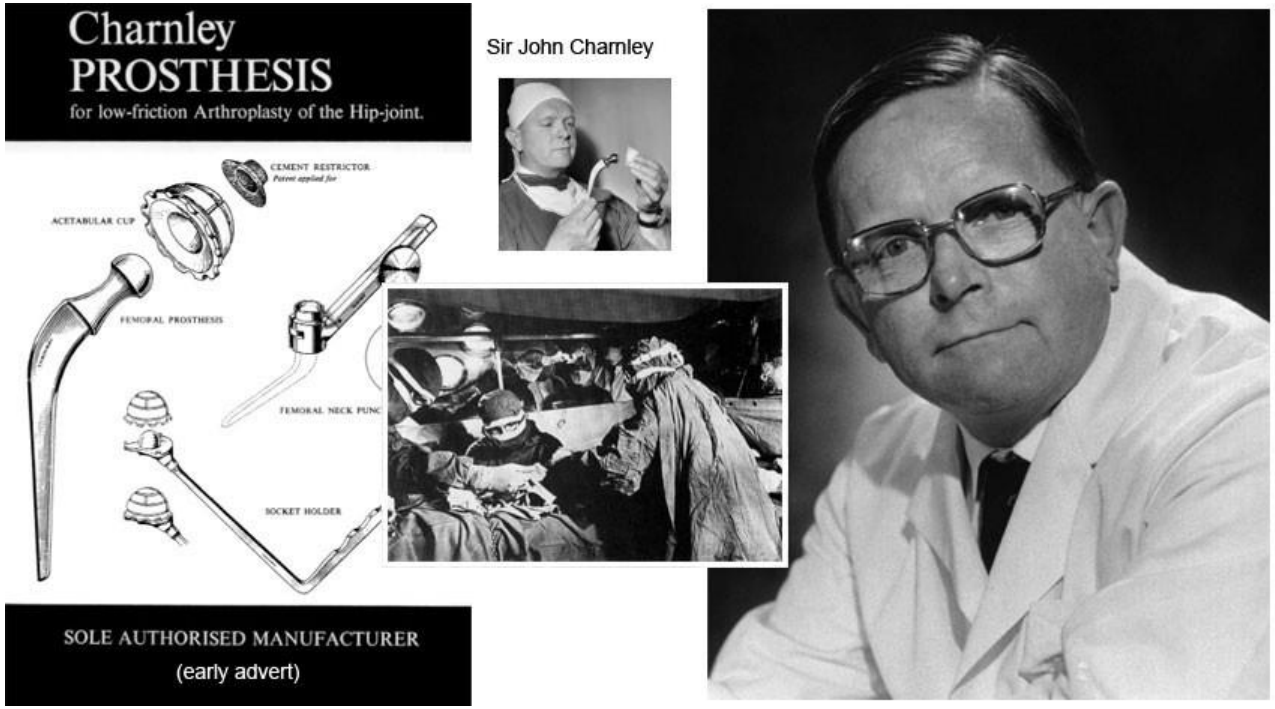


Hunter Mauston poses with his external pacemaker





1959: Sir John Charnley - Use of polymer in total joints

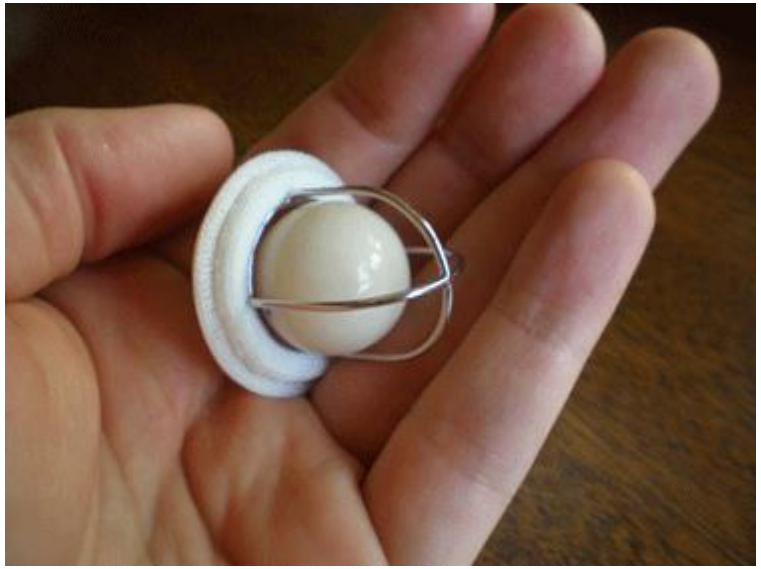


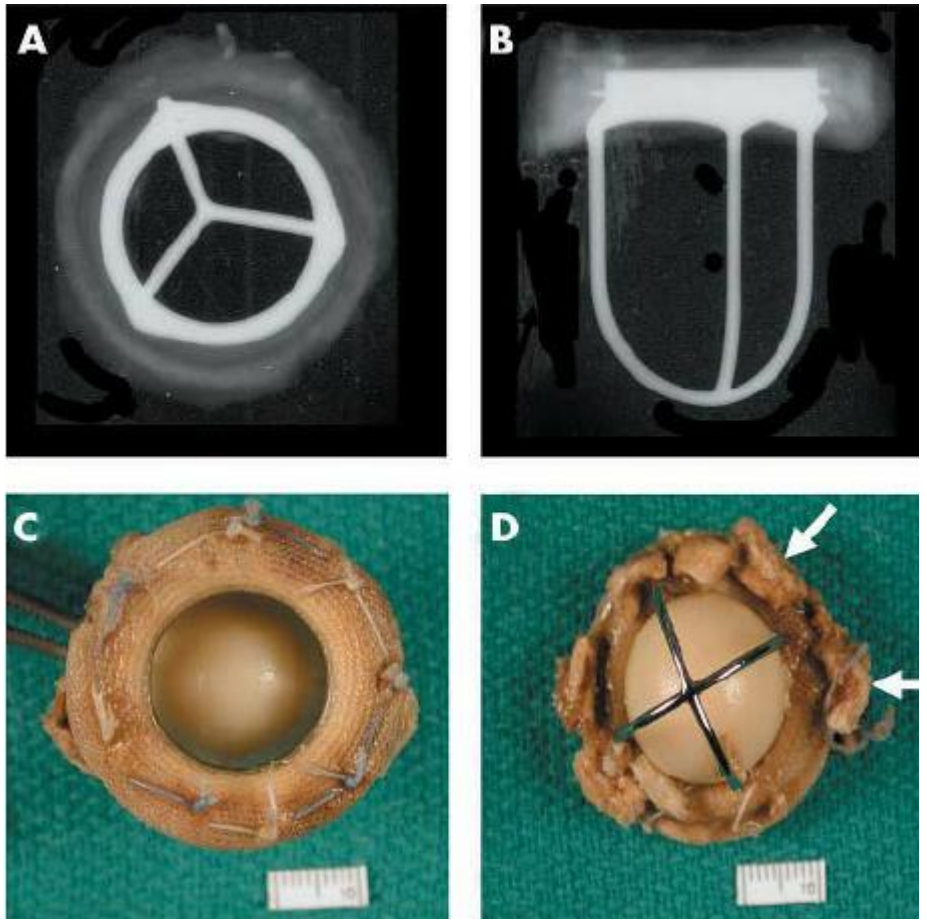






1960: L. Edwards, A. Starr Mitral valve replacement





1981: W. Kolff and others - Implantable artificial heart

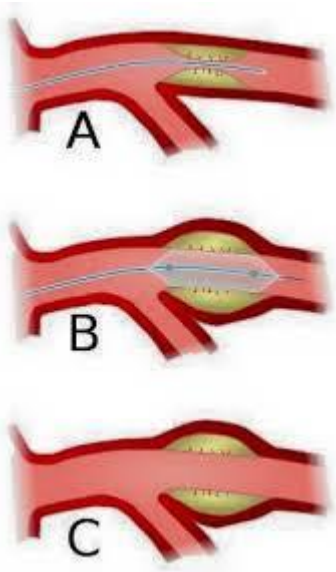


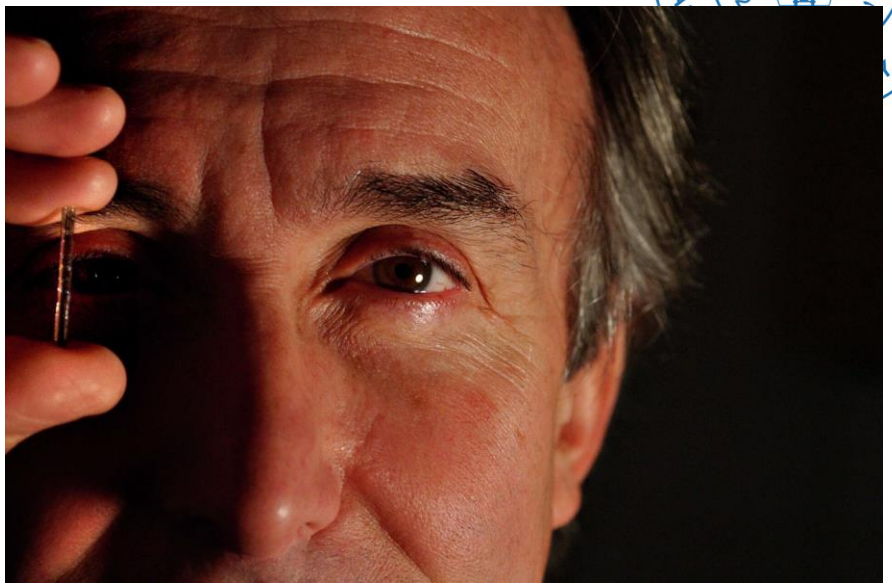




1980s: Julio Palmaz - Balloon expandable stent









Definitions

- Biomaterials do not necessarily have to be natural materials as the name may suggest.
- In 1974, at the 6th Annual International Biomaterials Symposium held at Clemson University, a biomaterial was defined as :
- https://blogs.clemson.edu/bioengineering/2018-clemson-awards-excerpted-from-a-match-a-history-of-the-society-for-biomaterials/

- ***a systemically, pharmacologically inert substance designed for implantation within or incorporation with a living system***

Definitions

- In 1986, at a consensus conference of the European Society for Biomaterials, a biomaterial was defined as
- https://catalog.nlm.nih.gov/discovery/fulldisplay/alma996473353406676/01NLM_INST:01NLM_INST
- *a nonviable material used in a medical device, intended to interact with biological systems.*
- *a material intended to interface with biological systems to evaluate, treat, augment, or replace any tissue, organ, or function of the body*

Definitions

- *a systemically, pharmacologically inert substance designed for implantation within or incorporation with a living system*

- *a nonviable material used in a medical device, intended to interact with biological systems.*

- *a material intended to interface with biological systems to evaluate, treat, augment, or replace any tissue, organ, or function of the body*

Biomaterials

- Biomaterials have unique applications in the body
- However, they belong to common material families (metals, polymers, ceramics, etc.)
- These same materials are used in other fields (construction, aerospace, sports)
- Therefore, biomaterials are not fundamentally different from other materials

Biomaterials

- Biomaterials definition and focus have evolved over time
- Early focus: mechanical properties (strength, stiffness, durability, corrosion resistance)
- Later: recognition of the importance of biological response
- Modern approach: design considers interaction with the body
- Current emphasis: controlling biological response using scientific principles
- Biomaterials are not just inert-they interact at cellular and molecular levels

Types of bonds in materials

- Materials can be classified as natural or synthetic
- More fundamentally, they are categorized by molecular structure and bonding type
- Bonding type strongly determines material properties
- Therefore, understanding different types of bonds is essential for biomaterials

Ionic bond

- Ionic bond: one atom donates an electron (cation), another accepts it (anion)
- Atoms are held together by electrostatic (Coulomb) attraction
- Example: NaCl → Na^+ and Cl^-
- Ions arrange to reduce repulsion (each ion surrounded by opposite charges)
- Bond is non-directional (equal strength in all directions)
- Forms highly ordered, strong, and stiff structures
- Materials are brittle (atoms cannot easily move)
- Poor electrical and thermal conductivity (electrons not free)
- High bond energy (600–1500 kJ/mol) → high melting points

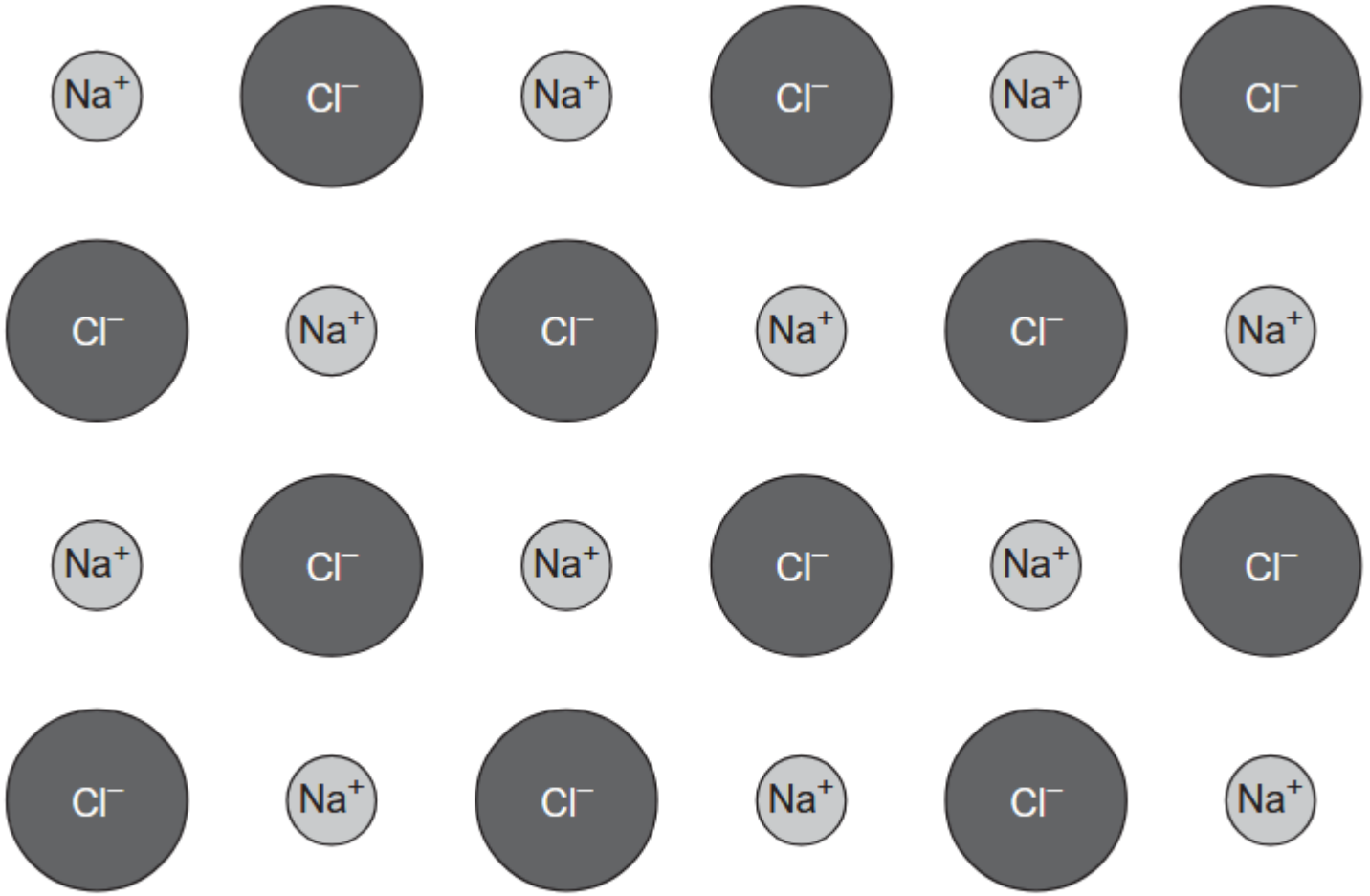


Figure 1.5

Ionic bonding between Na and Cl atoms.

Metallic bonds

- Metallic bond: metal atoms release valence electrons
- Forms positive ion cores surrounded by a free electron cloud
- Electron cloud acts like an adhesive, holding ions together
- Electrons are mobile → excellent electrical & thermal conductivity
- Bond is non-directional
- Layers of atoms can slide easily → metals are ductile and malleable
- Bond strength varies (e.g., mercury low, tungsten high)

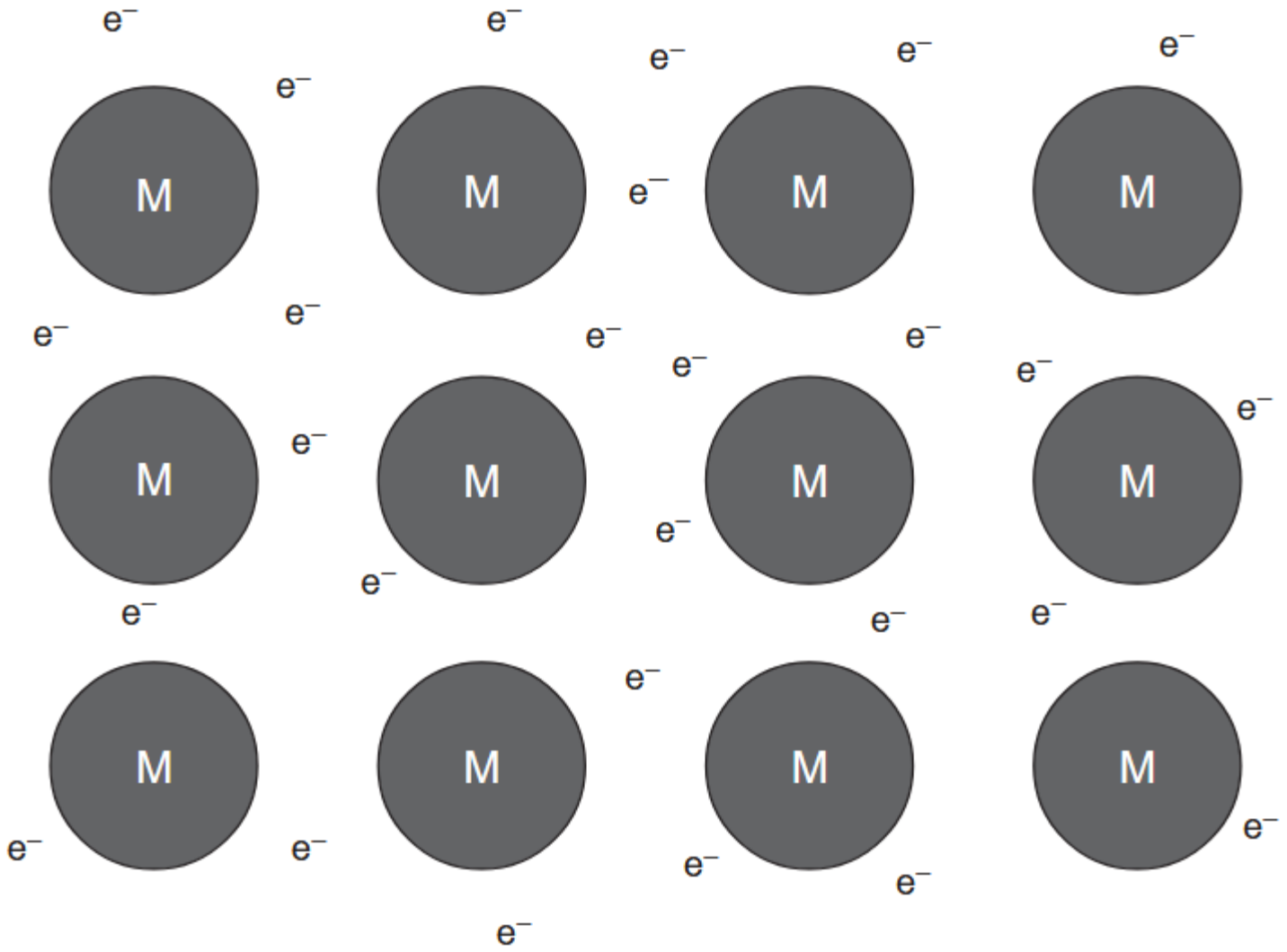


Figure 1.6

Metallic bonds showing positive ions surrounded by electron cloud.

Covalent bonds

- Covalent bond: atoms share electrons in outer shells
- Common in organic materials (e.g., ethylene, polyethylene) and non-metals (Cl_2, F_2)
- Examples: carbon-based materials, silicon carbide
- Bonds are strong and highly directional
- Single bonds allow rotation → gives flexibility
- Materials can have good deformability
- Electrons are not free → poor electrical & thermal conductivity

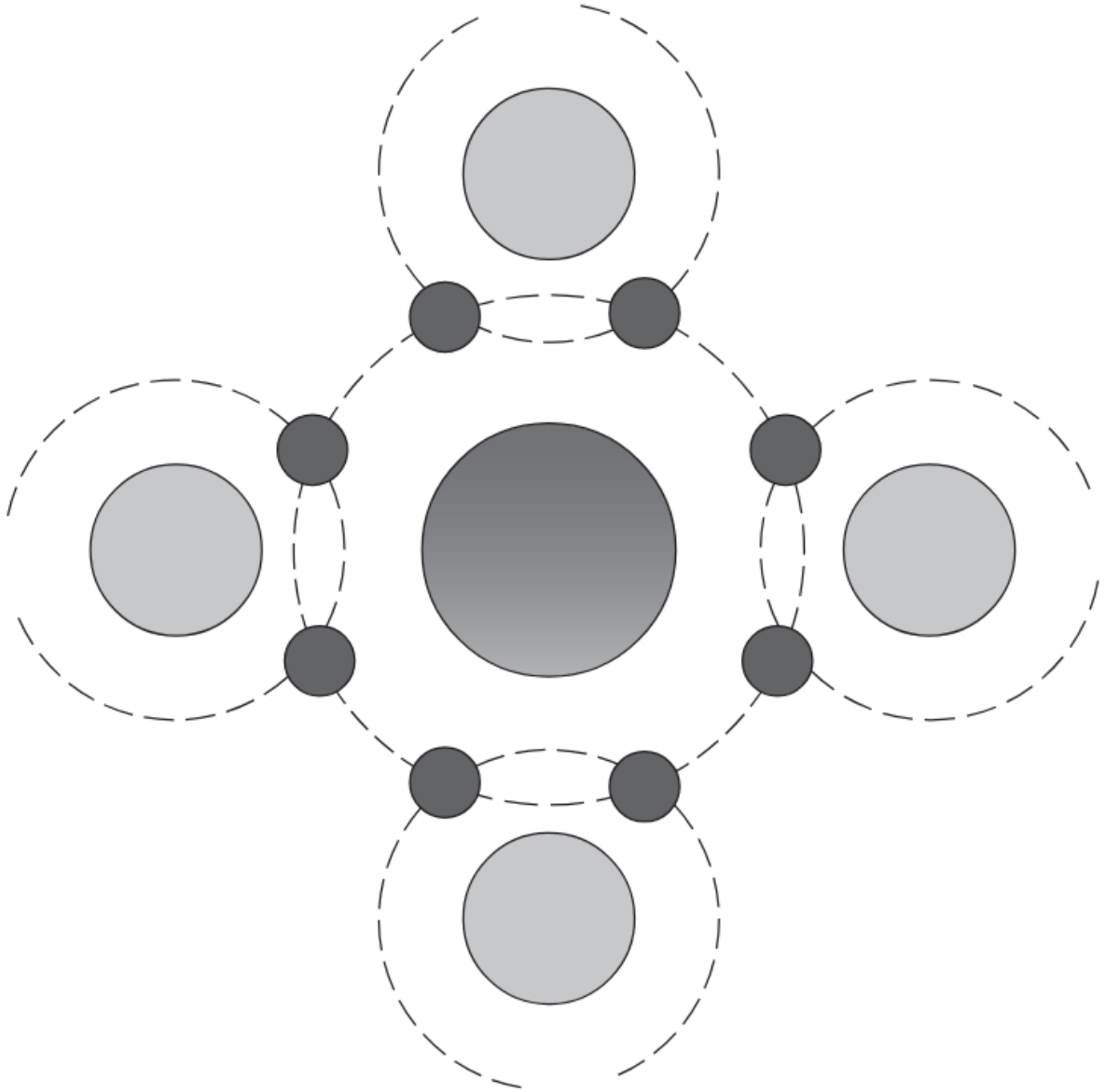


Figure 1.7

Covalent bonds showing atoms sharing electrons in their outer orbits.

Secondary bonds

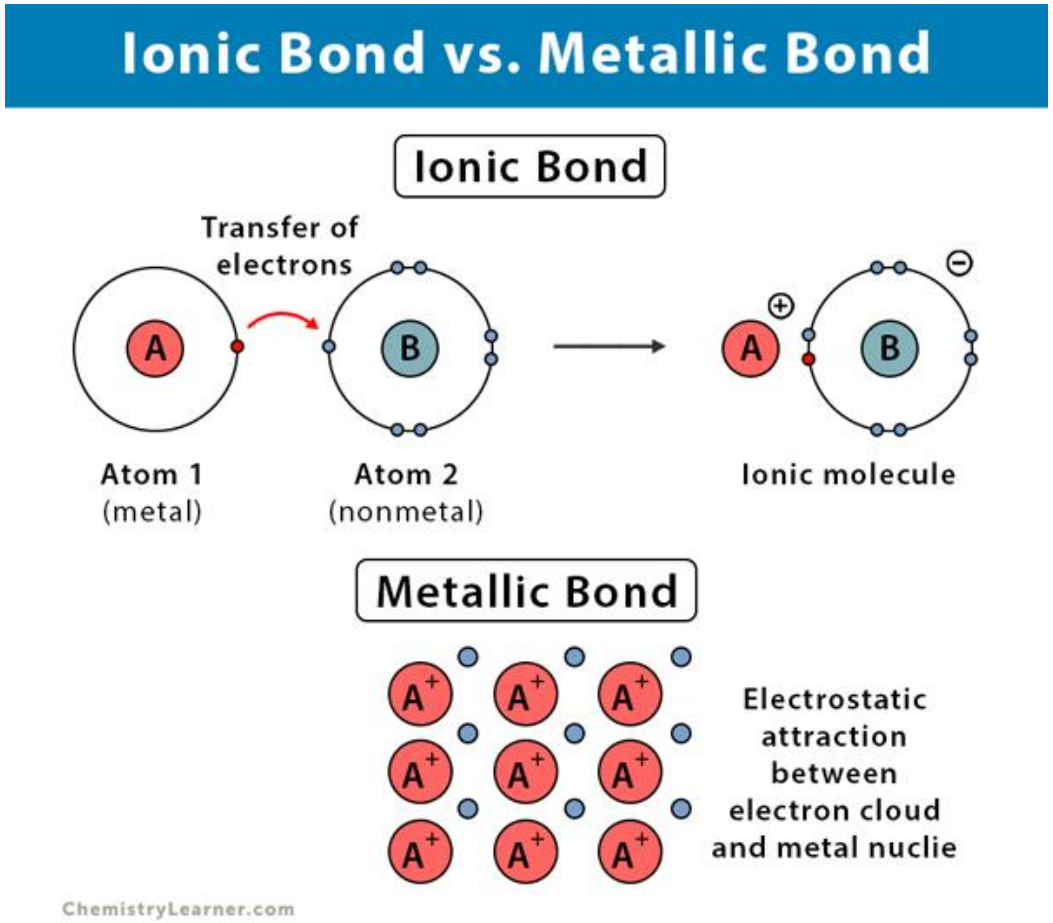
- Secondary bonds are weaker than primary bonds
- Include van der Waals and hydrogen bonds
- Based on attraction between dipoles (uneven charge distribution)
- Dipoles create electrostatic attraction between molecules
- Hydrogen bonds:
  - Occur when H is bonded to O, N, or F
  - Create strong polar molecules (dipoles)
  - Strongest secondary bond (~0.52 eV/atom)
- Other secondary bonds:
  - Weaker (~0.1 eV/atom)

Secondary bonds

- Secondary bonds do not involve electron sharing or transfer
- They are less directional than primary bonds
- Much weaker (<10% of covalent bond strength)
- Still significantly affect material properties, especially in polymers
- Understanding bonds helps in studying different material types

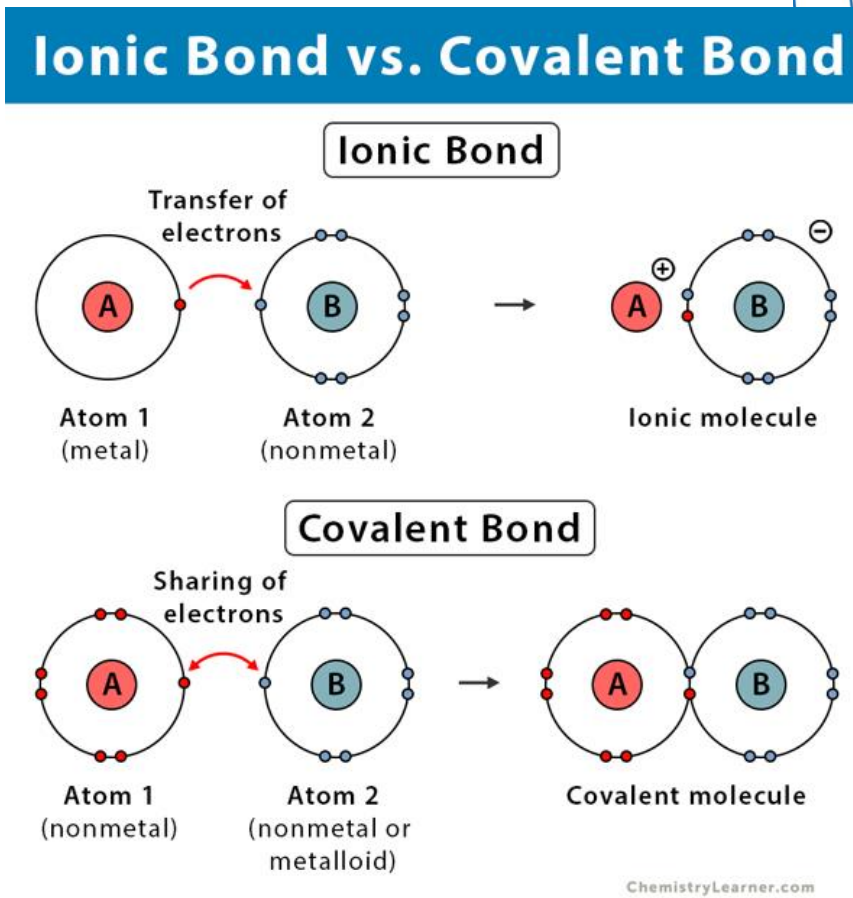
So →

- Bond strength order: Ionic > Metallic > Covalent > Secondary
- Ionic bonds:
  - Formed by electron transfer
  - Non-directional, uniform properties
  - Materials are strong but brittle
- Metallic bonds:
  - Electron cloud surrounds positive ion cores
  - Good heat & electrical conductivity



So →

- Covalent bonds:
  - Formed by electron sharing
  - Strong and directional
  - Poor conductors (electrons fixed)
- Secondary bonds:
  - Due to dipole attraction
  - Weakest bonds



Property	Ionic Bond	Metallic Bond	Covalent Bond	Secondary Bonds
Basic Mechanism	Electron transfer (cation + anion)	Electron cloud around metal ions	Electron sharing between atoms	Weak attraction between dipoles
Directionality	Non-directional	Non-directional	Highly directional	Weakly directional
Bond Strength	Very strong	Moderate to strong	Very strong	Weak
Structure	Ordered crystal lattice	Closely packed atoms	Specific molecular structures	Between molecules
Mechanical Behavior	Strong but **brittle**	**Ductile & malleable**	Strong, can be flexible	Weak, easily separated
Electrical Conductivity	Poor	Excellent	Poor	Very poor
Thermal Conductivity	Poor	Excellent	Poor	Very poor
Electron Mobility	Fixed (not free)	Free electrons	Fixed	No electron sharing
Examples	NaCl, ceramics	Metals (Fe, Ti, etc.)	Polymers, diamond, SiC	Polymers (van der Waals, H-bonds)
Role in Biomaterials	Ceramics (bone substitutes, implants)	Load-bearing implants (orthopedic, dental)	Polymers & organic tissues	Important in polymer behavior

Types of materials

Types of materials: Ceramics

- Derived from “keramos” (Greek) → related to heat/burning processes
- Made of ionic or mixed ionic–covalent bonds
- Typically compounds of metal + non-metal (oxides, carbides, nitrides)
- Can be crystalline or amorphous (glass)





Types of materials : Ceramics – Mechanical Properties

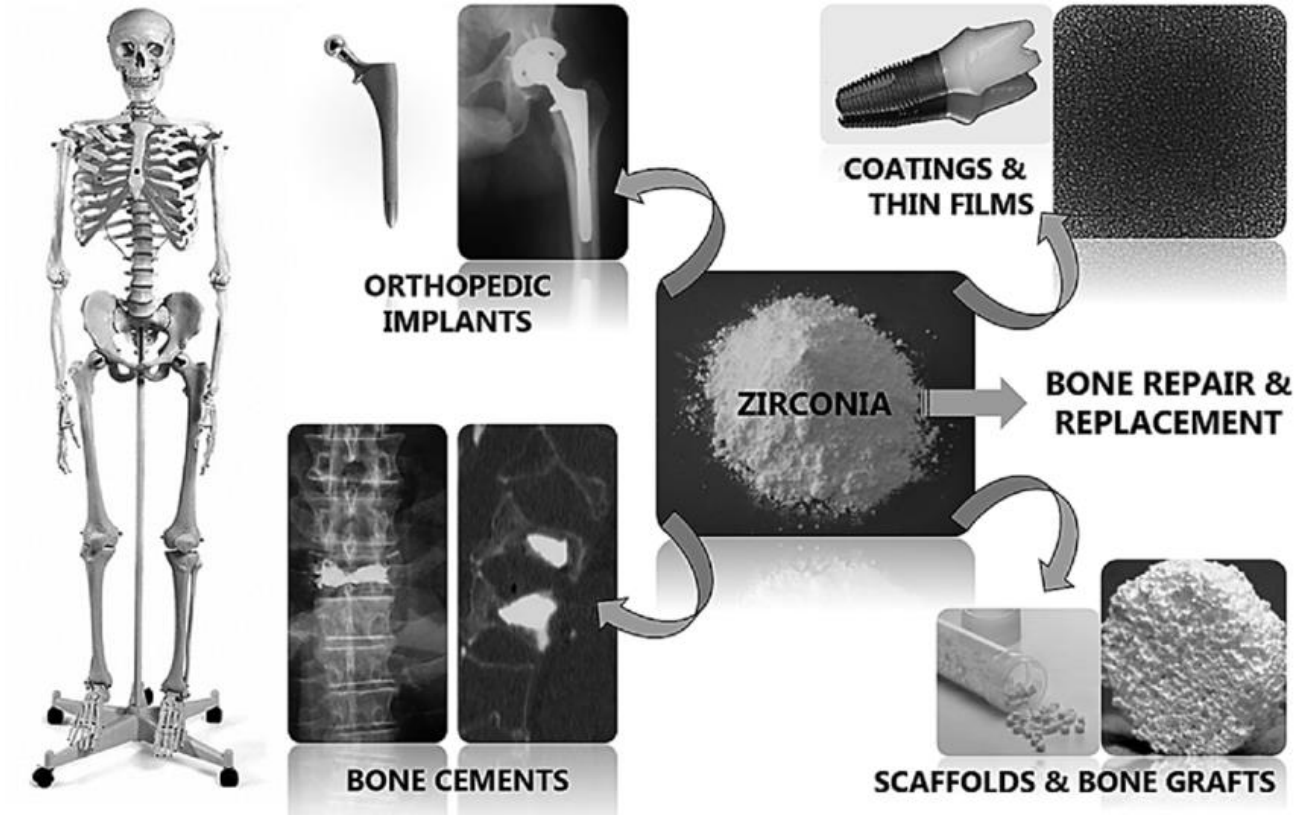
- Very hard and stiff (high elastic modulus)
- High compressive strength
- Low ductility → brittle
- Low tensile strength & fracture toughness
- Sensitive to cracks (stress concentration → failure)
- Poor resistance to crack propagation

Types of materials : Ceramics - Thermal , Electrical & other Properties

- Thermal & Electrical Properties
  - Good insulators (heat & electricity)
  - High temperature resistance
  - Excellent chemical stability
- Other Properties
  - Can have good wear resistance (when polished)
  - In fiber form → improved mechanical properties (used in composites)

Types of materials : Ceramics - Biomedical Applications

- Joint prostheses
- Bone/tissue scaffolds (orthopedics)
- Dental crowns
- Heart valves



***Zirconium dioxide (ZrO_2)*

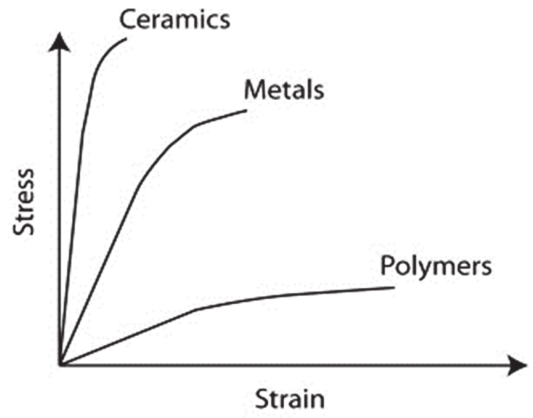
Types of materials: Metals

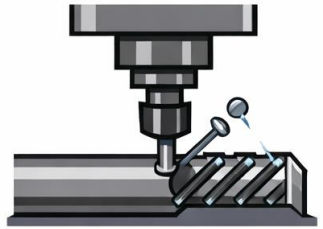
- Inorganic materials with non-directional metallic bonds
- Atoms are closely packed → high density, visible in X-rays
- Surrounded by free electron cloud



Types of materials: Metals - Mechanical Properties

- High strength & elastic modulus
- High melting point
- Ductile → can be shaped easily
- Can be processed by machining, forging, casting, forming





Machining → Remove

Cutting Away Material



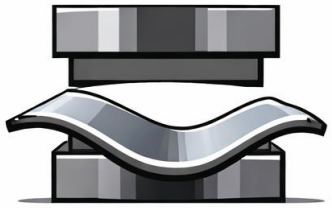
Forging → Hit / Press

Hammer & Press Metal



Casting → Pour

Pouring Molten Metal

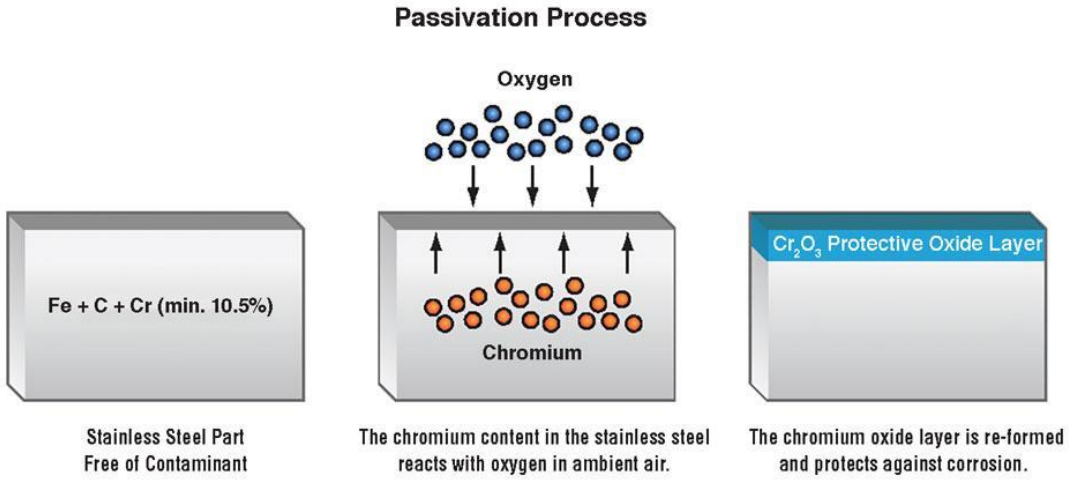


Forming → Reshape

Bending & Shaping Metal

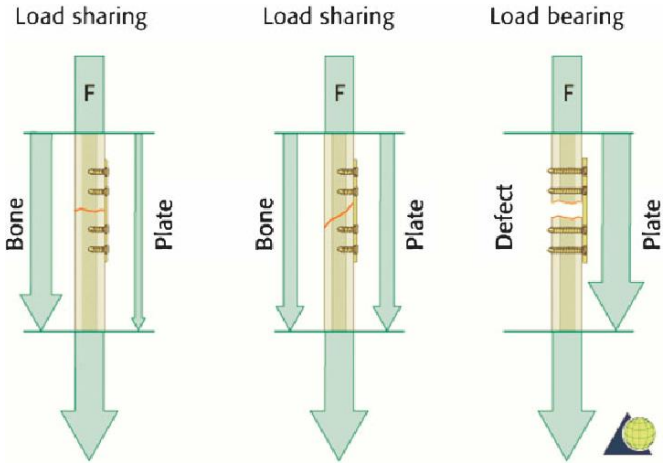
Types of materials: Metals - Thermal & Electrical Properties and Limitations

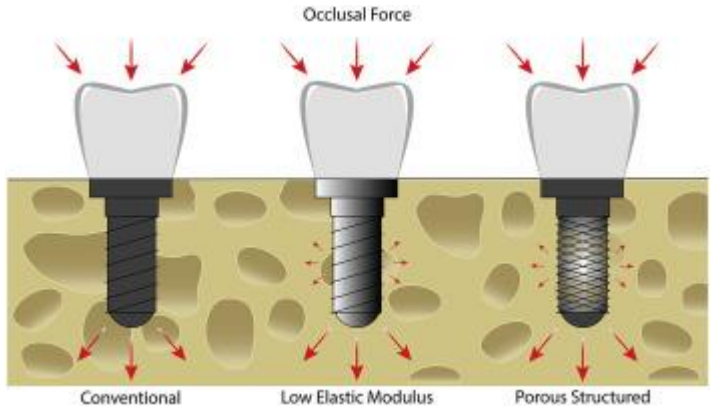
- Excellent conductors of heat and electricity
- Susceptible to corrosion (especially in chloride environments)
- Require passivation for biomedical use (Passivation = protection layer forms on metal surface)



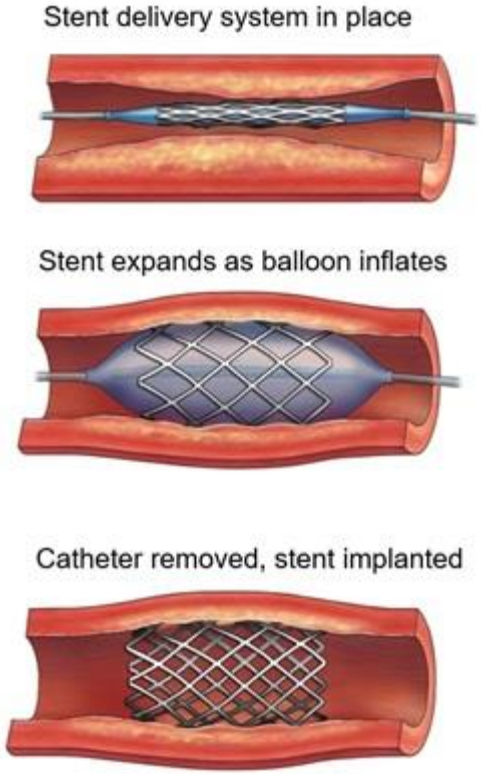
Types of materials: Metals - Biomedical Applications

- Load-bearing implants (orthopedic, dental)
- Fixation devices
- Also used in stents, guide wires, heart valves, electrodes
- Metals have the longest history in biomaterials



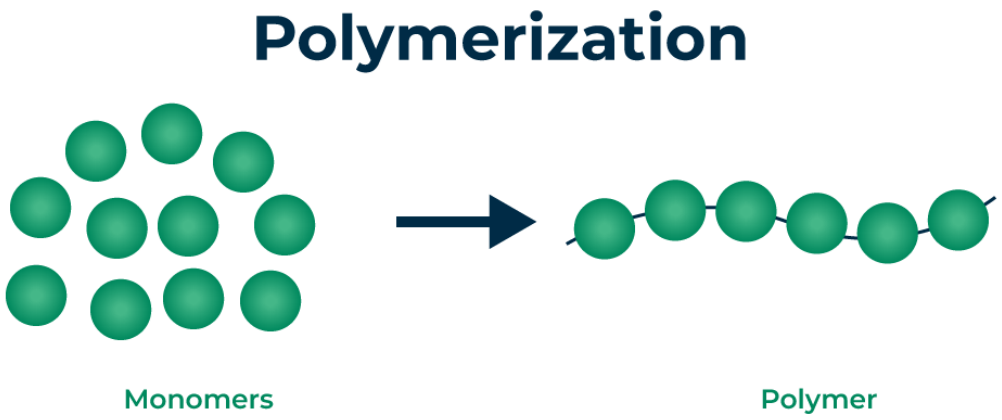


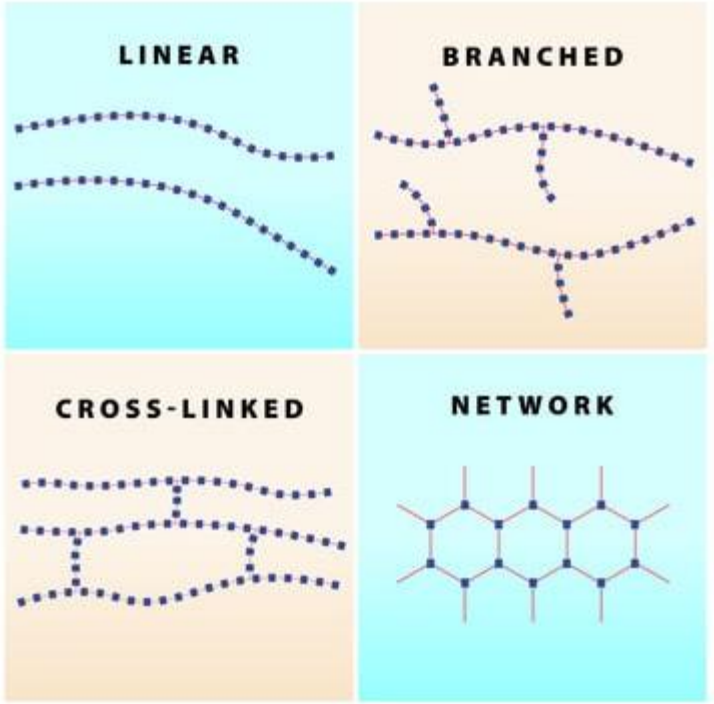




Types of materials: Polymers

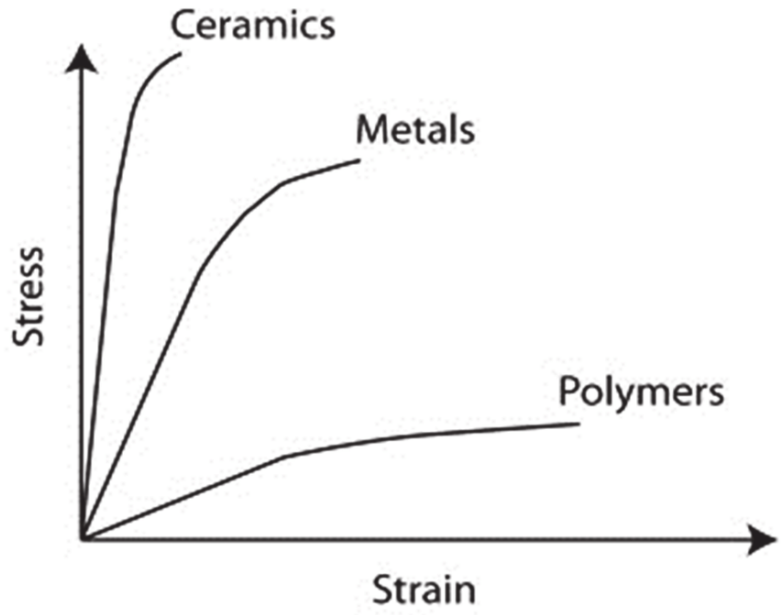
- Organic materials with long chains linked by covalent bonds
- Mainly composed of C, H, and other non-metals
- Chains vary in length, can be entangled or cross-linked
- Interactions via covalent, van der Waals, or hydro
- Structure: crystalline, amorphous, or mixed





Types of materials: Polymers - Mechanical Properties

- Flexible, ductile, highly deformable
- Chains can slide or rotate (unless cross-linked)
- Low strength, elastic modulus, and thermal resistance (vs metals/ceramics)

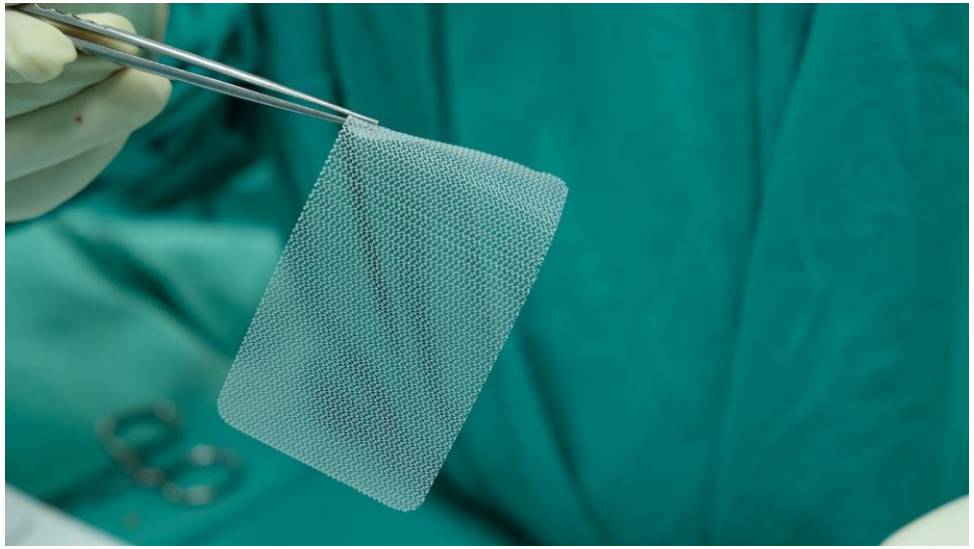


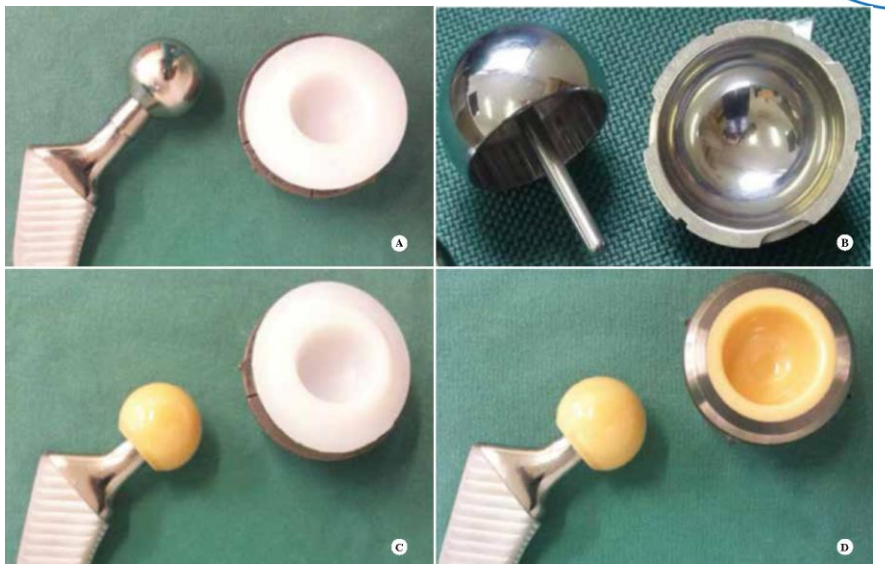
Types of materials: Polymers - Thermal , Electrical Properties and Manufacturing

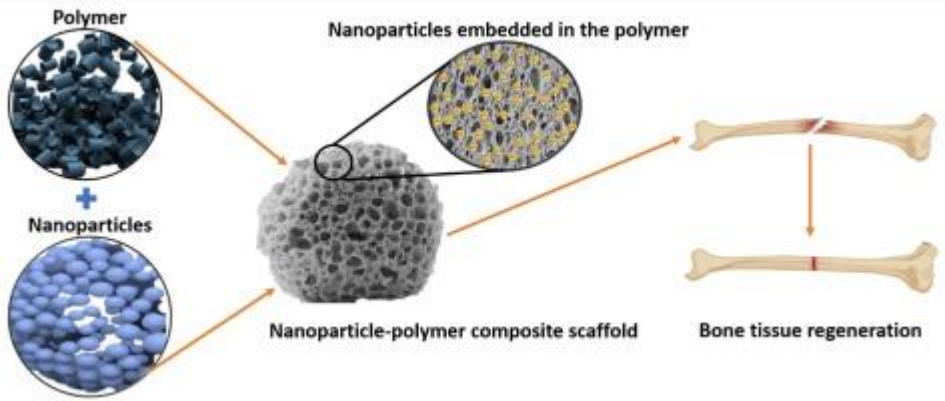
- Poor conductors of heat and electricity
- Easy and low-cost production
- Shaped by molding techniques (e.g., melt molding)
- Increasing importance in biomaterials over the last 50 years

Types of materials: Polymers - Biomedical Applications

- Sutures, coatings, pacemakers
- Joint bearing surfaces
- Resorbable meshes, plates, screws
- Contact lenses, tissue scaffolds

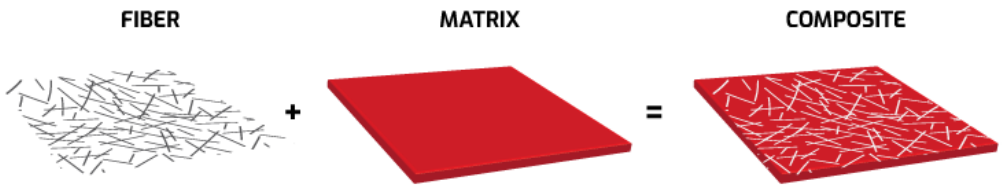


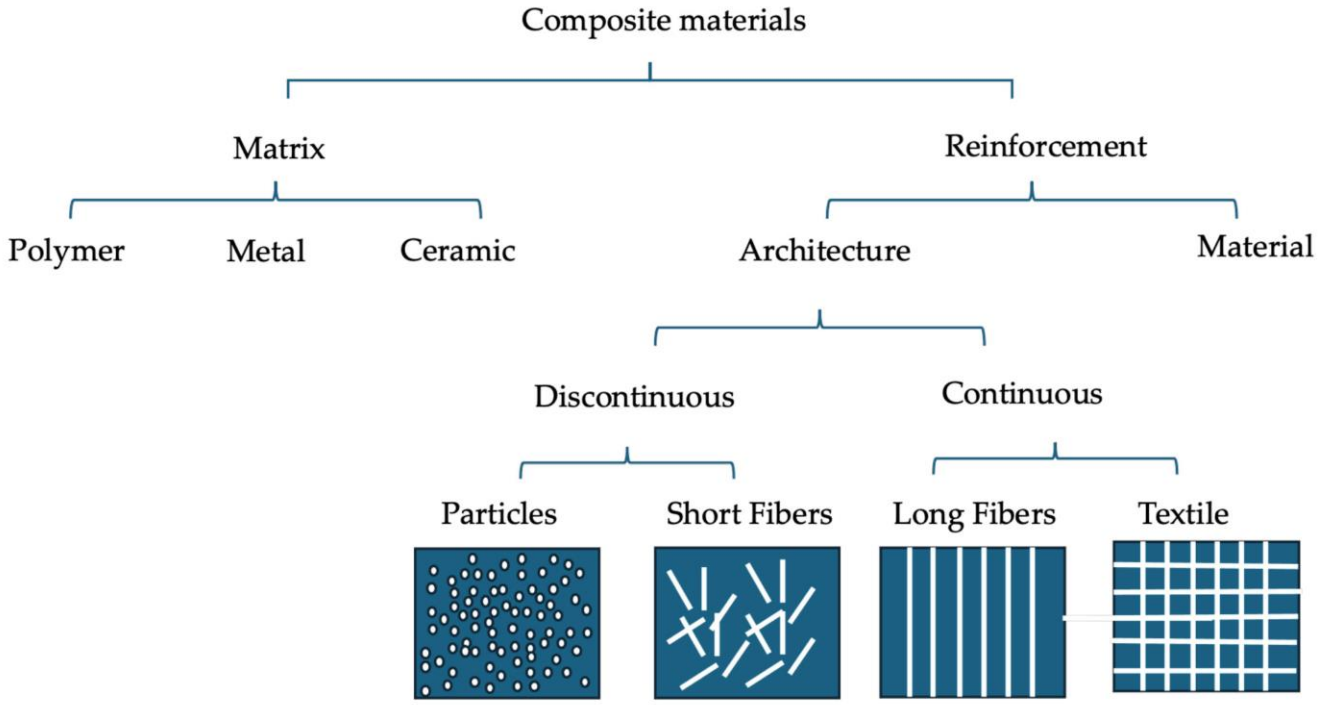




Types of materials: Composites

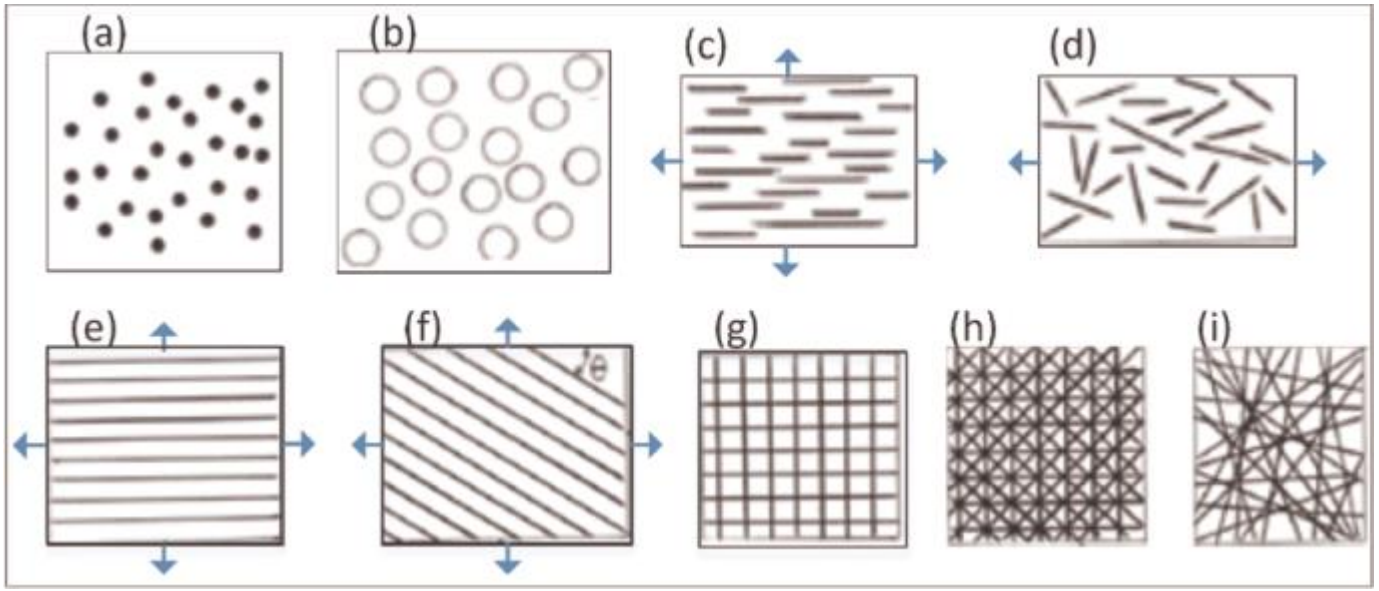
- Made of two or more distinct materials (phases) with clear interfaces
- Consist of matrix + filler (reinforcement)
- Usually combine different materials





Types of materials: Composites - Properties Depend On

- Shape, size, orientation, distribution of components
- Relative proportions of each phase



Types of materials: Composites - Mechanical Behavior

- Often use strong fibers in a polymer matrix → high strength & toughness
- Fiber-reinforced composites → can be anisotropic (direction-dependent)
- Particulate composites → usually isotropic

Examples of Biomaterials & Their Applications

Table 1.2 Examples of biomaterials and their applications

Biomaterial	Applications
Ceramics	
Alumina, Zirconia	Total joint prostheses, dental implants, implant coatings
Calcium phosphates	Tissue regeneration scaffolds, cements, drug delivery systems
Metals	
Stainless steel	Fracture fixation devices, stents, wires, dental implants
Cobalt chrome alloy	Total joint prostheses, stents
Titanium and alloys	Total joint prostheses, dental implants
Nitinol	Stents
Polymers (synthetic)	
Polyethylene	Total joints for hips, knees etc.
Polymethylmethacrylate	Bone cement
Polytetrafluoroethylene	Vascular grafts
Polyethylene terephthalate	Vascular grafts, hernia repair meshes
Polydimethyl siloxane	Contact lenses, breast implants
Polyhydroxyethyl methacrylate	Contact lenses
Polyurethane	Coatings for long term implants
Polylactic acid	Resorbable screws
Poly(lactic-co-glycolic) acid	Resorbable sutures
Polycaprolactone	Drug delivery devices
Polymers (natural)	
Collagen	Tissue engineering scaffolds, cosmetic bulking agent
Chitosan	Wound dressings

Impact of Biomaterials

- Biomaterials are essential in modern healthcare, helping improve and save lives.
- Applications
  - Orthopedics: joint replacements, fracture fixation
  - Dentistry: dental implants
  - Eye care: contact lenses
  - Cardiovascular: stents
  - Medical devices: pacemakers, heart valves
- Biomaterials are a core part of healthcare and their use is rapidly increasing

Future Trends in Biomaterials

- Rapid advancement due to improved understanding of biology at cellular and molecular levels
- Increasing focus on interactions between biomaterials, proteins, and cells that control the body's response
- Recognition that implants are quickly covered by biological molecules after insertion
- Strong trend toward hybrid and surface-modified biomaterials
- Use of biological components (e.g., proteins, peptides) to enhance integration and long-term performance

Think with me 🤔🤔🤔

- 1.What is the main difference between the old definitions for biomaterials and those from more recent years?
- 2. Which of the following would be considered to be made from a biomaterial according to the modern definitions for a biomaterial: (a) pacemaker for the heart, (b) walking stick, (c) stethoscope, (d) vascular graft, (e) toothbrush, (f) suture?



Thank you